DETAIL SPECIFICATION SHEET

# HOSE ASSEMBLY, RUBBER, LIGHTWEIGHT, MEDIUM PRESSURE, INTERNAL SUPPORT COIL, FIELD ATTACHABLE END FITTINGS, FLARELESS TO FLANGE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-83796.

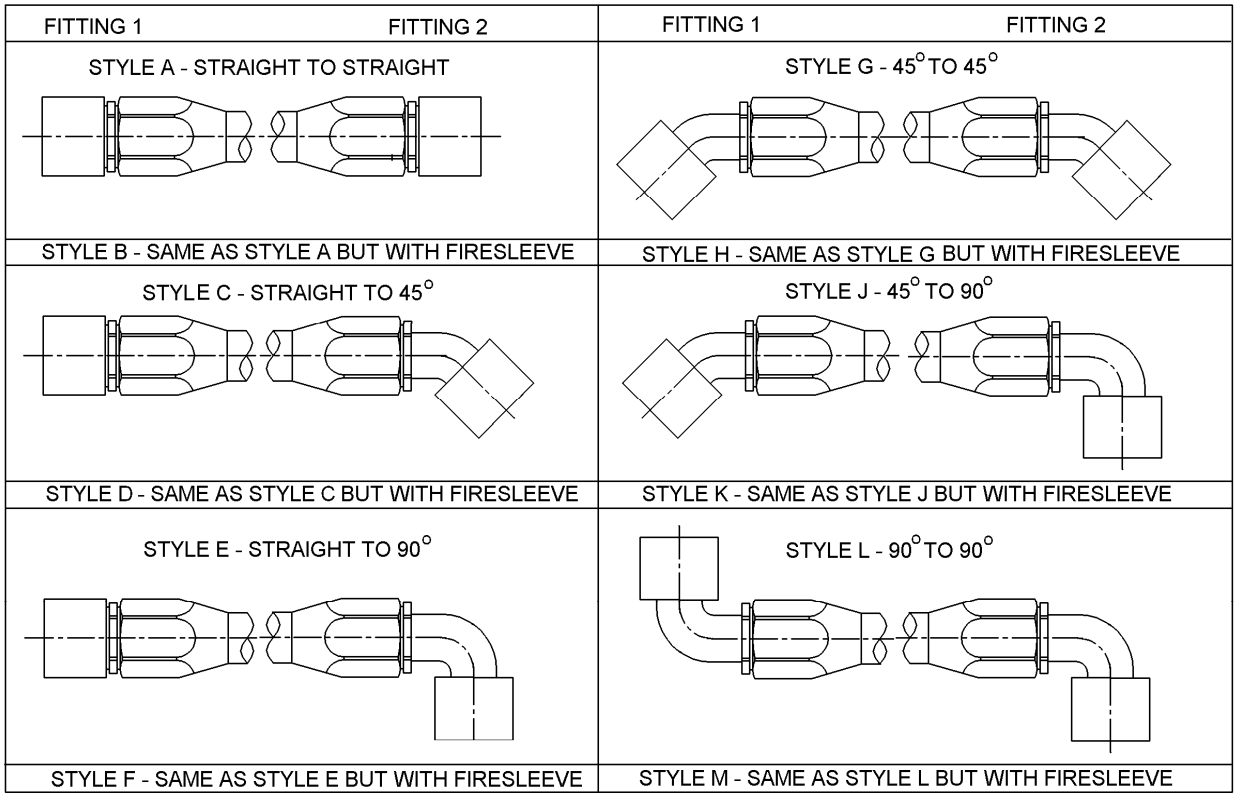



FIGURE 1. Hose assembly styles, field attachable fittings, flare-to-flare.



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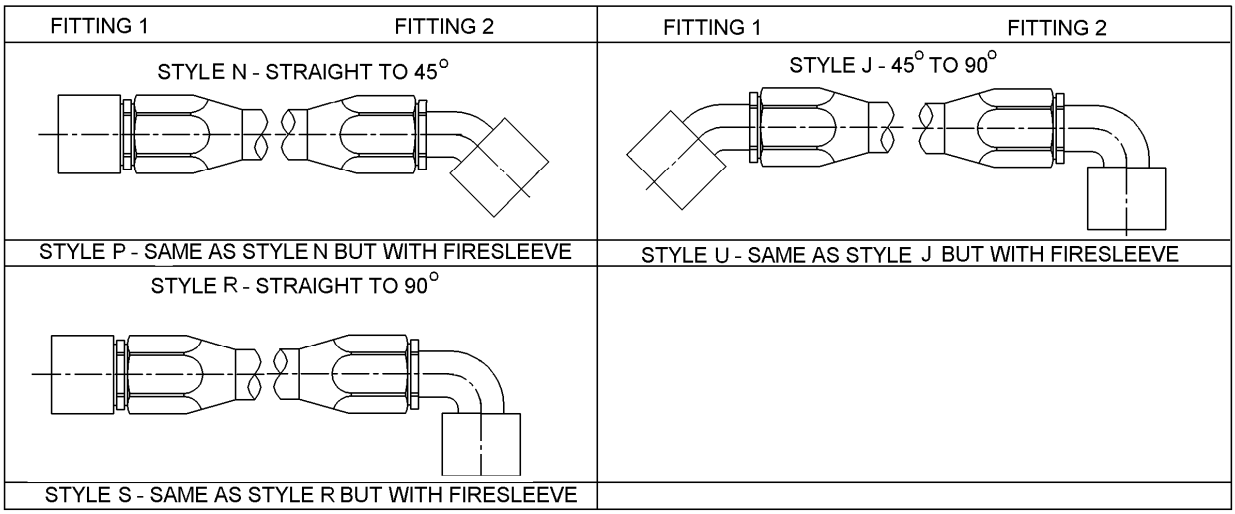



FIGURE 1. Hose assembly styles, field attachable fittings, flare-to-flare - Continued.

Hose assembly elbow fitting drop height. Elbow fitting drop height, A and B as shown on figure 2, are as specified in the applicable fitting specification sheet (see table I).

TABLE I. Hose assembly, field attachable fittings, HCOF, styles A thru M.

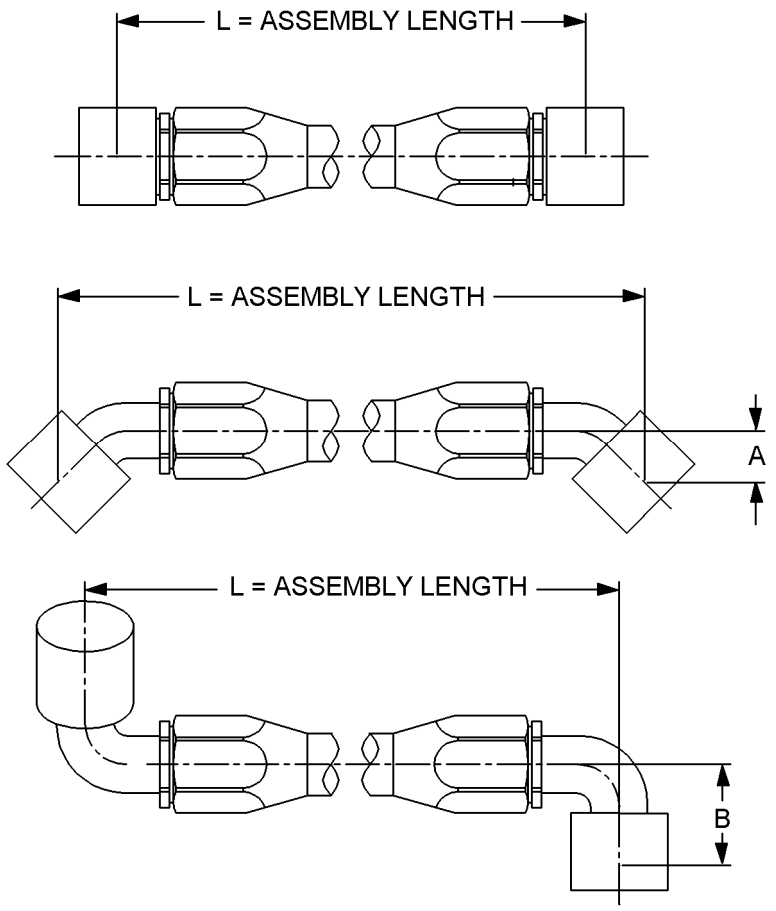
| Style | Fitting ends MIL-DTL-83798 | | MIL-DTL-83797 hose cut off factor (HCOF) 1/ inches verses hose dash size number | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | Fitting 1 | Fitting 2 | -08 | -10 | -12 | -16 | -20 | -24 | -32 |
| A and B | M83798/4 | M83798/7 | 2.34 | 2.42 | 2.62 | 3.06 | 3.57 | 3.67 | 4.20 |
| C and D | M83798/4 | M83798/8 | 2.41 | 2.59 | 2.94 | 3.47 | 3.91 | 4.36 | 5.03 |
| E and F | M83798/4 | M83798/9 | 2.34 | 2.55 | 2.96 | 3.53 | 4.03 | 4.54 | 5.32 |
| G and H | M83798/5 | M83798/8 | 2.69 | 2.94 | 3.47 | 4.09 | 4.61 | 5.11 | 5.82 |
| J and K | M83798/5 | M83798/9 | 2.62 | 2.90 | 3.49 | 4.15 | 4.73 | 5.29 | 6.11 |
| L and M | M83798/6 | M83798/9 | 2.30 | 2.54 | 3.18 | 3.86 | 4.46 | 4.94 | 5.72 |
| N and P | M83798/5 | M83798/7 | 2.62 | 2.77 | 3.15 | 3.48 | 4.17 | 4.42 | 4.69 |
| R and S | M83798/6 | M83798/7 | 2.30 | 2.41 | 2.84 | 3.39 | 3.90 | 4.07 | 4.60 |
| T and U | M83798/6 | M83798/8 | 2.37 | 2.58 | 3.16 | 3.80 | 4.34 | 4.76 | 6.43 |

1/ Example of hose cut-off factor: for a -6, 3/8 hose OD, 20-1/2 assembly length, style A, the bulk hose length = 20.50 - the HCOF = 20.50 -2.08 = 18.42. Bulk hose length required is 18.44 inches.

Hose assembly length. The hose assembly length is calculated as shown on figure 2.



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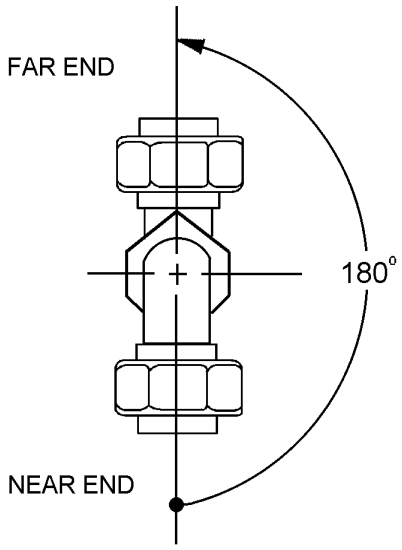

NOTE: Length “L” to be measured between centers of nipple ends along a straight line parallel to hose length with hose laid out horizontally and straight.

FIGURE 2. Hose assembly dimensions.



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Angular displacement. Angular displacement shall be as shown on figure 3.




NOTES:

1. Angular displacement for hose assemblies with elbow fittings on both ends shall have the angular displacement between elbows, measured counter-clockwise from the centerline of the nearest fitting, positioned at six-o' clock, to the centerline of the other fitting (see SAE-J517).
2. The near end of the connector shall be put in numerical order relative to the far end (see table I). Example: Near end adapter 45°, far end 90°.
3. Angular displacement shall be measured in degrees with a tolerance of ± 3° for lengths up to 2 feet (61 cm) and ± 5° for all lengths over 2 feet (61 cm).
4. The angular displacement shall be expressed in two digits number of degrees divided.
5. Making the angular determination in the wrong direction will result in an unacceptable part.

FIGURE 3. Elbow fitting drop height and angular alignment.

REQUIREMENTS

Hose assemblies shall be specified on figures 1, 2, and 3.

Fittings shall mate with parts in accordance with SAE-AS4395 and the mounting pad in accordance with MS33786, as applicable.

Hose assembly length tolerance shall be in accordance with MIL-DTL-83796.

Internal support coil. Internal support coil shall be in accordance with Defense Supply Center Columbus drawing 08001.

Internal support coil length. The internal support coil length shall be the overall hose assembly length minus the sum of the overall lengths of each end fitting.

Size codes shall be as specified in table II. Flange fittings not available for sizes -03, -04, -05, and -06.



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TABLE II. Size code. 1/ 2/

| Dash number | | -08 | -10 | -12 | -16 | -20 | -24 | -32 |
|---|---|---|---|---|---|---|---|---|
| Size code | | H | J | K | M | N | P | R |
| Hose OD reference | Inches fraction | 1/2 | 5/8 | 3/4 | 1 | 1 1/4 | 1 1/2 | 2 |
| | Inches decimal | .500 | .625 | .750 | 1.000 | 1.250 | 1.500 | 2.000 |
| | mm | 12.70 | 15.88 | 19.05 | 25.40 | 31.75 | 38.10 | 50.80 |

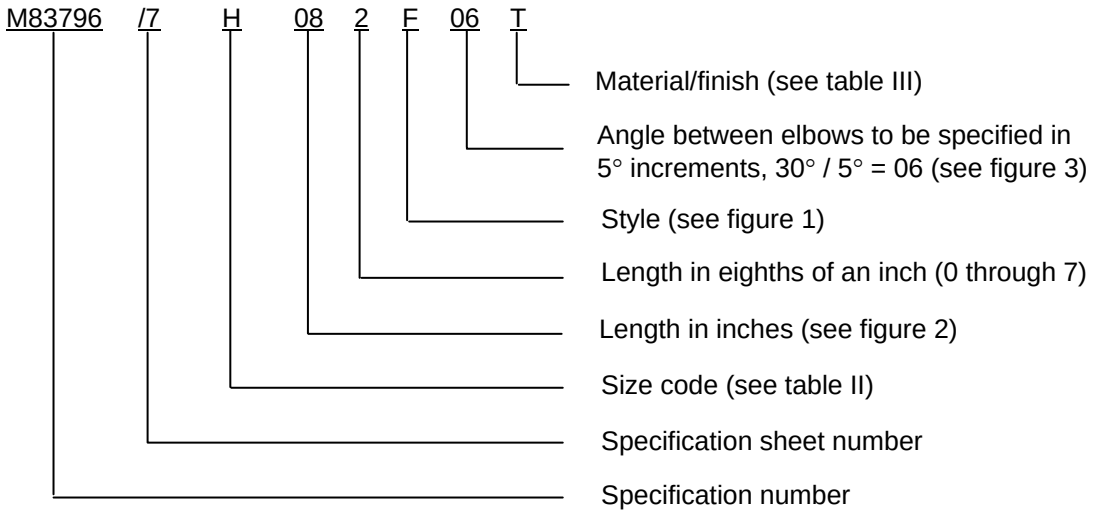
1/ Dimensions are in inches.
2/ Metric equivalents are given for information only.

Materials and finishes shall be in accordance with MIL-DTL-83796 and in table III.

TABLE III. Material and finish code.

| Part or Identifying Number (PIN) code dash letter | Material | Plating finish |
|---|---|---|
| Blank | Aluminum | Anodize in accordance with MIL-A-8625, type II |
| -T | Titanium (all sizes) | Annealed |

PIN example:




M83796/7H082F06 indicates - Hose assembly for 1/2 hose OD, 8 1/4 inches in length, style F, 30° between elbows, titanium alloy.



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Referenced documents. In addition to MIL-DTL-83796, this document references the following:

MIL-A-8625
MIL-DTL-83797
MIL-DTL-83798
08001
MS33786
SAE-AS4395
SAE-J517

## CONCLUDING MATERIAL

Custodians:
Army - AV
Navy - AS
Air Force - 99
DLA - CC

Preparing activity:
DLA - CC

(Project 4720-2007-064)

Review activities:
Army - AT
Navy - MC, SA
Air force - 71, 85

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.



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